
Course Name: Machine Learning for Earth System Sciences

Week 02 - Assignment 02 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

In a Geophysical Network two events A and B with time series V_i and V_j respectively is said to be synchronised at time t if,

- a) $|t_A - t_B| < \text{threshold}$
- b) $|t_A - t_B| > \text{threshold}$
- c) $|t_A - t_B| = \text{threshold}$
- d) $|t_A - t_B| \geq \text{threshold}$

Answer: (a)

Detailed Solution:

In a Geophysical Network two events are said to be synchronised if the absolute value of their difference is less than the threshold value.

QUESTION 2

In the context of climate networks, geophysical/geographical networks help identify:

- I. Regions whose climatic or geophysical conditions are strongly related.
- II. Teleconnections between geographically distant regions.
- III. Causal relationships between geophysical events.

Which of the above statements is/are true?

- a) I only
- b) I and II only
- c) II and III only
- d) I, II and III

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Answer: (b)

Detailed Solution:

A climate network edge indicates that two locations exhibit a significant statistical relationship, for example:

- high correlation,
- mutual information,
- event synchronization.

It does not imply that one location or event causes the other.

QUESTION 3

In a Geophysical Network the probability that two randomly chosen neighbours of any node are themselves neighbour is called as

- a) Local/Global Clustering Coefficient
- b) Degree Distribution
- c) Centrality
- d) Area weighted connectivity

Answer: (a)

Detailed Solution:

Local Clustering Coefficient measures the probability that two neighbours of a node are also connected to each other, indicating the tendency of the node to form a local cluster. In Climate Networks it is used to identify localized regions where climate variables (e.g., rainfall or temperature) exhibit strong interconnected behaviour, such as regional monsoon systems or coherent weather patterns.

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QUESTION 4

Identify the True Statement:

- a) When the cause and effect relate to the same time period then it is called contemporaneous causality.
- b) Granger causality is suitable for detecting contemporaneous causality.
- c) Granger causality cannot be bidirectional.
- d) None of these.

Answer: (a)

Detailed Solution:

When the cause and effect relate to the same time period then it is called contemporaneous causality. The Granger causality is based on the premises that (i) a cause occurs before its effect and (ii) knowledge of a cause improves prediction of its effect. Hence Granger causality is not suitable for detecting contemporaneous causality. Also, Granger causality can be bidirectional.

QUESTION 5

Which of the following statements is incorrect about Data Assimilation?

- a) It is a technique whereby observational data are combined with output from a numerical model to produce an optimal estimate of the evolving state of the system
- b) Data assimilation techniques like Ensemble Kalman Filter and 4D-Var have the limitation that they can handle only linear systems.
- c) The observations are used to estimate the evolving state of the system and, in some applications, calibrate model parameters.
- d) It is used to keep updating a dynamical model with observations.

Answer: (b)

Detailed Solution:

Earth systems—such as atmospheric fluids and ocean currents—are notoriously highly nonlinear and chaotic (governed by non-linear partial differential equations like the Navier-Stokes equations). If data assimilation only worked for linear systems, modern weather forecasting would be impossible. Advanced DA techniques like the Ensemble Kalman Filter (EnKF) and 4D-Var (Four-Dimensional Variational Data

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Assimilation) were explicitly designed to handle the severe nonlinearities of geoscientific systems.

QUESTION 6

The estimate of the length of a box is updated using the following Kalman Filter equation:

$$\hat{x}_k = \hat{x}_{k-1} + 0.5(z_k - \hat{x}_{k-1}),$$

where the Kalman Gain is constant ($K=0.5$).

Initially, $\hat{x}_0 = 6.0$ inches. The measured lengths are:

Reading No.	1	2	3	4
Measure Length (in cms)	6.2	6.6	6.1	7.1

What is the estimate after the 4th update?

- a) 6.66 inches
- b) 6.71 inches
- c) 6.76 inches
- d) 7.10 inches

Answer: (a)

Detailed Solution:

Update 1 : $6.0 + 0.5(6.2 - 6.0) = 6.1$

Update 2: $6.1 + 0.5(6.6 - 6.1) = 6.35$

Update 3: $6.35 + 0.5(6.1 - 6.35) = 6.225$

Update 4: $6.225 + 0.5(7.1 - 6.225) = 6.6625$

≈ 6.66 inches.

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QUESTION 7

Suppose the occurrence of heavy rainfall in a city follows a stationary stochastic process. If the probability of heavy rainfall on day t is 0.25, then the probability of heavy rainfall on which of the following days is also expected to be 0.25?

- a) $t+1$
- b) $t+2$
- c) $t+3$
- d) All of the above

Answer: (d)

Detailed Solution:

For a stationary stochastic process, the probability structure does not change over time, so if the probability of heavy rainfall on day t is 0.25, it is also expected to be 0.25 on $t+1$, $t+2$, and $t+3$.

QUESTION 8

Which of the following is/are true about causality?

- I. Granger causality can't be misled by confounders.
- II. The standard linear Granger non-causality test is effective only when time series are stationary
- III. The linear Granger causality test is implemented by fitting autoregressive models.

- a) I only
- b) I and II
- c) I and III
- d) II and III

Answer: (d)

Detailed Solution:

Standard linear Granger Causality is implemented by autoregressive models and is applicable only for stationary time series. In case of non-stationary data, vector auto

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regressive models are used. Confounding variables are an issue for Granger's Causality

QUESTION 9

Which of the following statements is/are true

- I. Frechet, Weibull and Gumbel distributions are examples of extreme value distributions.
- II. Gamma distribution has an exponential right-hand tail.
- III. Anomaly events exhibit spatial coherence but can never exhibit temporal coherence.

- a) I only
- b) I and II
- c) I and III
- d) II and III

Answer: (b)

Detailed Solution:

Anomaly events can exhibit both spatial and temporal coherence. Examples:

- Spatial coherence: A heatwave may simultaneously affect neighbouring regions.
- Temporal coherence: A heatwave or drought may persist over several consecutive days or weeks.

QUESTION 10

A weather forecasting model uses 100 predictor variables. The researcher wants the model to automatically identify and remove irrelevant variables. Which regression technique is most suitable?

- a) Linear Regression
- b) Ridge Regression
- c) Lasso Regression
- d) Logistic Regression



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Answer: (c)

Detailed Solution:

Lasso Regression is designed for variable selection because it can shrink irrelevant coefficients to zero, effectively removing unimportant predictors from the model.